

AQI Value Prediction Using Deep Learning

Team Information

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1. Introduction

Air pollution has become one of the most serious environmental challenges worldwide. Poor air quality affects human health, visibility, climate, and overall quality of life. The Air Quality Index (AQI) is a numerical indicator used to communicate how polluted the air is and its potential health impacts.

This project aims to develop a Deep Learning Regression Model capable of predicting AQI values using pollutant concentrations and meteorological parameters. Accurate AQI prediction can help authorities and citizens take preventive measures against harmful pollution levels.

2. Problem Statement

The objective of this project is to predict the numerical AQI value of a city based on environmental and weather-related factors. Instead of classifying AQI into categories, the model predicts the exact AQI value, making it a regression problem.

3. Dataset Description

The dataset contains air quality and weather information collected from multiple cities.

Input Features

- City
- PM2.5
- PM10
- NO₂
- SO₂
- CO
- O₃
- Temperature
- Humidity
- Wind Speed
- Atmospheric Pressure

Target Variable

- AQI (Air Quality Index)

The City feature is transformed using One-Hot Encoding so that it can be used effectively by machine learning and deep learning algorithms.

4. Methodology

The following steps were performed during project development:

Data Preprocessing

1. Handling missing values.
2. Removing duplicate records.
3. One-Hot Encoding of city names.
4. Feature scaling using StandardScaler.
5. Splitting data into training and testing sets.

Model Development

A Deep Neural Network (DNN) was built using TensorFlow/Keras.

Model Architecture:

- Input Layer
- Hidden Layer (ReLU)
- Hidden Layer (ReLU)
- Output Layer (Linear Activation)

The model was trained using the Adam optimizer and Mean Squared Error (MSE) loss function.

5. Evaluation Metrics

The performance of the regression model was evaluated using:

Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted AQI values.

Mean Squared Error (MSE)

Measures the average squared prediction error.

Root Mean Squared Error (RMSE)

Provides error values in the same unit as AQI.

R² Score

Indicates how well the model explains the variance in AQI values.

6. Expected Results

The model is expected to learn the relationship between pollutant concentrations, weather conditions, and AQI values. Pollutants such as PM2.5 and PM10 are anticipated to have the highest influence on AQI prediction.

The trained model should provide accurate AQI estimates and assist in understanding environmental pollution trends.

7. Applications

- Smart City Monitoring Systems
 - Environmental Risk Assessment
 - Public Health Advisory Systems
 - Pollution Forecasting
 - Government Air Quality Monitoring Programs
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8. Conclusion

This project demonstrates the use of Deep Learning for environmental data analysis. By predicting AQI values using pollutant and meteorological data, the proposed model can support air quality monitoring and decision-making processes. The project highlights the practical application of Artificial Intelligence in solving real-world environmental challenges and promoting sustainable development.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-Learn
- TensorFlow / Keras
- Matplotlib
- Seaborn
- Jupyter Notebook